

Smart City Civic Complaint and Management System

Software Engineering Project Journal – 02

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Journal 02

Requirements Analysis and System Modelling

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1. Objective

The objective of this journal is to document the requirements analysis and system modelling activities carried out for the Smart City Civic Complaint and Management System.

The main objectives were:

- To understand the functional requirements of the proposed civic complaint management system.
- To identify the major actors involved in the system and understand their responsibilities.
- To analyze the major use cases associated with citizens, department officers, field workers, and administrators.
- To understand the complete lifecycle of a civic complaint from submission to resolution and closure.
- To study the Use Case Diagram and Activity Diagram prepared for the project.
- To review the project Gantt Chart and understand the planned development schedule.
- To organize and understand the existing project artifacts before proceeding with implementation.

2. Work Completed

Following the completion of the project proposal, the next stage involved understanding the functional requirements and modelling the proposed system.

The project requirements and existing Software Engineering artifacts were studied to obtain a clearer understanding of the system before implementation.

The major activities covered in this journal were:

1. Requirements analysis
2. Identification of system actors
3. Identification of functional requirements
4. Analysis of major use cases
5. Study of the complaint management workflow
6. Review of the Use Case Diagram
7. Review of the Activity Diagram

8. Review of the project Gantt Chart
9. Organization of project artifacts in the GitHub repository

These activities provide a foundation for the implementation and testing stages of the project.

3. System Overview

The Smart City Civic Complaint and Management System is a web-based platform designed to provide a structured mechanism for citizens to report civic issues and track their resolution.

The system connects citizens with department officers and field workers through a centralized complaint management workflow.

A complaint can contain information such as:

- Complaint category
- Description of the problem
- Location
- Photograph or other supporting evidence

After submission, the complaint is reviewed by a department officer. Once verified, it can be assigned to a field worker for resolution. The field worker performs the required work, updates progress, and uploads evidence of the completed work.

The citizen can subsequently verify the resolution and provide feedback. If the issue has not been properly resolved, the complaint can be reopened.

This creates a traceable workflow from complaint creation to final resolution.

4. System Actors

The system consists of four major user roles.

Actor	Main Responsibilities
Citizen	Register or log in, submit complaints, provide complaint details and location, upload photographs, track complaint status, verify resolution, provide feedback, and reopen complaints when required.
Department Officer	Review submitted complaints, verify complaints, assign field workers, monitor progress, and manage complaint status.

Field Worker	View assigned tasks, accept work, travel to the complaint location, perform the required work, update progress, upload before/after evidence, and mark the work as completed.
Administrator	Manage users, departments, and workers, monitor complaints, monitor SLA-related information, and view system-level analytics.

The identification of these actors helps define the responsibilities and access requirements of different users within the system.

5. Functional Requirements

The major functional requirements identified for the system are divided according to the responsibilities of each user role.

5.1 Citizen Requirements

The citizen should be able to:

- Register and log in to the system.
- Submit a new civic complaint.
- Select an appropriate complaint category.
- Enter a description of the issue.
- Provide the location of the complaint.
- Upload a photograph as supporting evidence.
- Receive a unique complaint ID.
- Track the status of a submitted complaint.
- Verify whether the reported issue has been resolved.
- Provide feedback or rating after resolution.
- Reopen a complaint when the issue has not been resolved satisfactorily.

5.2 Department Officer Requirements

The department officer should be able to:

- View newly submitted complaints.
- Review complaint details, photographs, and location.
- Verify whether a complaint is valid.
- Reject invalid complaints with an appropriate reason.
- Assign verified complaints to field workers.
- Monitor work progress.
- Update complaint status when required.

5.3 Field Worker Requirements

The field worker should be able to:

- View assigned tasks.
- Accept an assigned task.
- View the complaint location and details.
- Update work progress.
- Upload before and after photographs as evidence.
- Mark the assigned work as completed.

5.4 Administrator Requirements

The administrator should be able to:

- Manage system users.
- Manage departments.
- Manage field workers.
- Monitor complaints.
- Monitor service-level information.
- View analytics related to complaints and system activity.

6. Use Case Analysis

The Use Case Diagram was analyzed to understand the interaction between the system and its external actors.

The major use cases identified for the system include:

- Register / Login
- Submit Complaint
- Track Complaint Status
- Verify Resolution
- Provide Feedback
- Reopen Complaint
- Select Category
- Enter Description
- Provide Location
- Generate Complaint ID
- Upload Photograph
- Review Complaint
- Verify Complaint
- Assign Worker
- Monitor Progress
- Manage Complaint Status
- View Assigned Tasks
- Accept Work
- Update Work Progress
- Upload Evidence
- Mark Work Completed
- Manage Users
- Manage Departments

- Manage Workers
- Monitor Complaints

The Use Case Diagram also includes the planned Smart Assistance layer. This layer contains:

- Smart Classification
- Duplicate Detection
- Priority Calculation

These smart features are intended to improve the efficiency of complaint processing after the core complaint workflow has been established.

7. Use Case Diagram

The Use Case Diagram provides a functional view of the proposed system by showing the major actors and their interactions with the system.

The diagram distinguishes the responsibilities of citizens, department officers, field workers, and administrators.

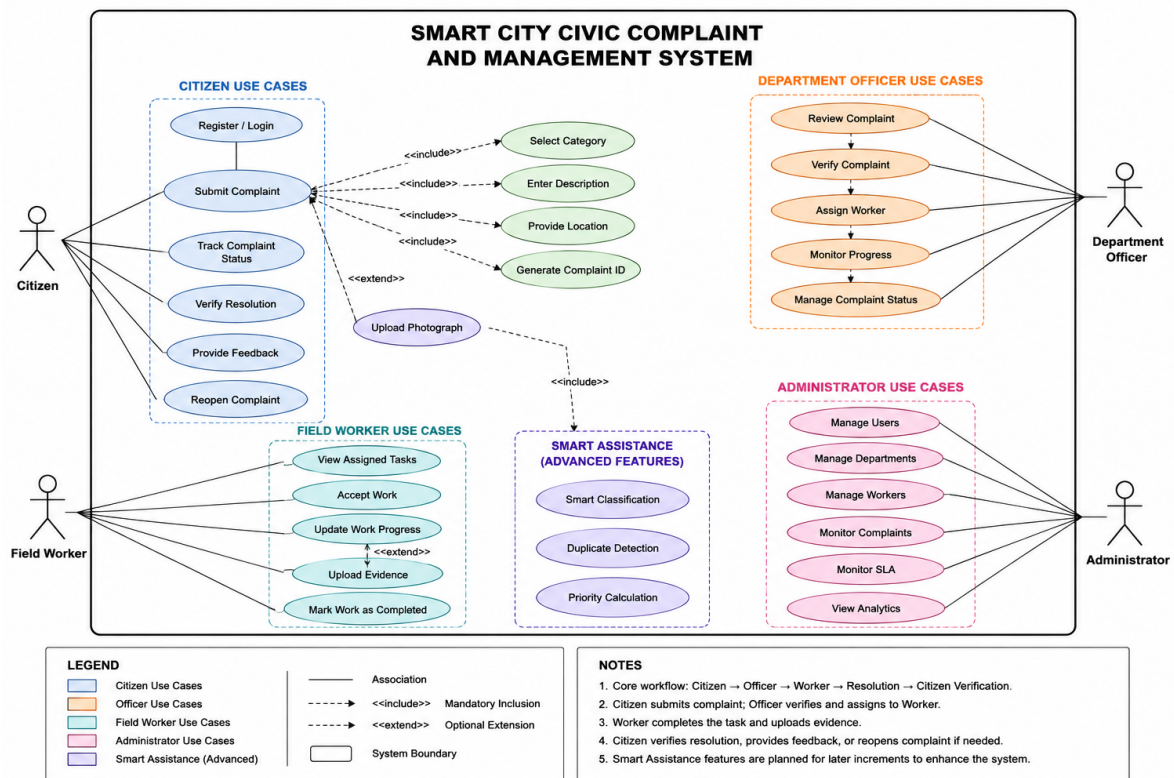


Figure 1: Use Case Diagram of the Smart City Civic Complaint and Management System

The diagram helps establish the system boundary and provides a reference for the functionality that will be implemented during development.

8. Activity Diagram Analysis

The Activity Diagram was studied to understand the sequence of activities involved in processing a complaint.

The major stages of the workflow are:

1. Citizen registration or login.
2. Submission of a civic complaint.
3. Validation of the submitted information.
4. Generation of a complaint ID.
5. Storage of the complaint in the database.
6. Setting the complaint status to “Submitted”.
7. Notification of the department officer.
8. Review and verification of the complaint.
9. Assignment of a field worker.
10. Field worker acceptance of the task.
11. Execution of the required work.
12. Updating of work progress.
13. Uploading of before and after evidence.
14. Marking the work as completed.
15. Updating the complaint status.
16. Notification of the citizen.
17. Citizen verification of the resolution.
18. Closure of the complaint if the issue is resolved.
19. Reopening of the complaint if the issue remains unresolved.

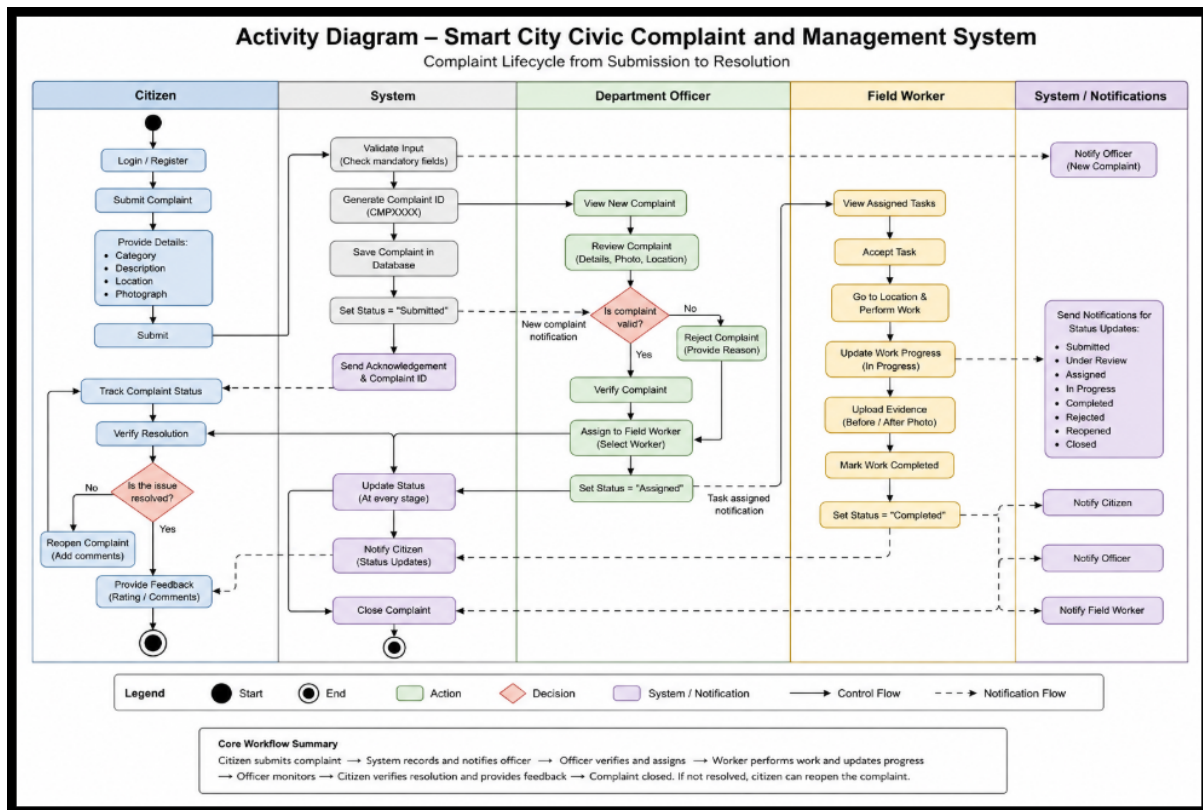


Figure 2: Activity Diagram of the Complaint Lifecycle

The activity diagram provides a process-oriented view of the system and complements the functional information represented in the Use Case Diagram.

9. Core Complaint Workflow

The complete complaint lifecycle identified during the analysis can be represented as:

Citizen → Complaint Submission → Officer Verification → Worker Assignment → Work and Evidence → Feedback

The workflow provides traceability throughout the complaint lifecycle.

If the complaint is valid, the department officer verifies it and assigns it to an appropriate field worker.

The field worker then performs the required work and uploads evidence. After the work is completed, the citizen is notified and can verify the resolution.

If the citizen is not satisfied with the resolution, the complaint can be reopened for further action.

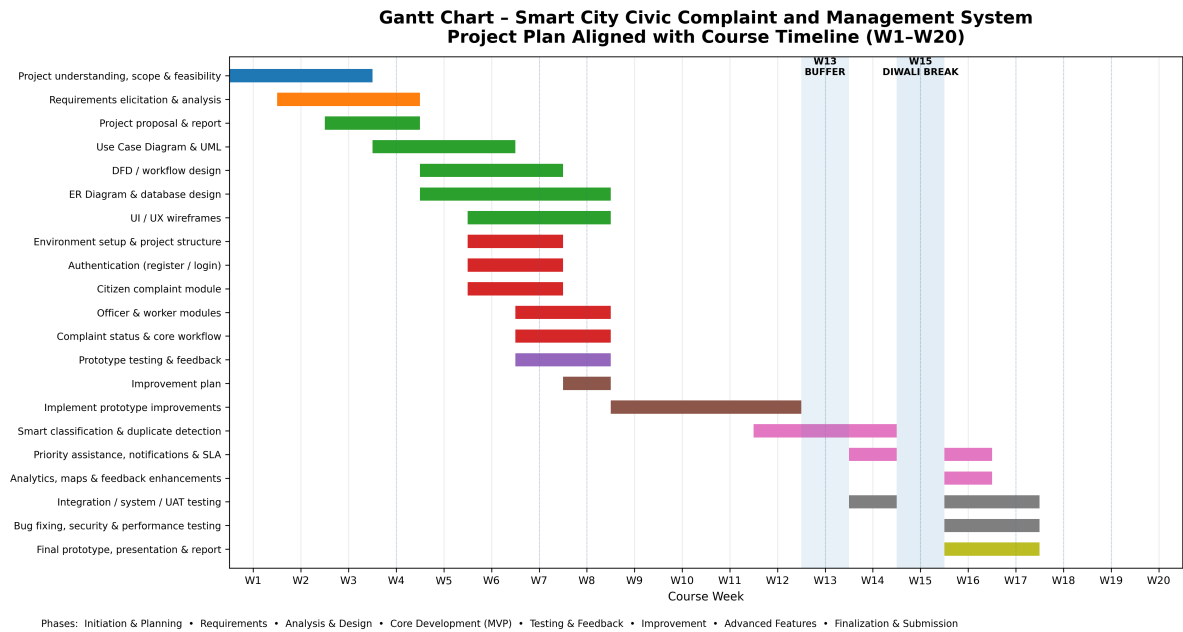
10. Project Schedule

The project Gantt Chart was reviewed to understand the planned development schedule.

The project follows an incremental development approach in which the core complaint workflow is prioritized before implementing advanced smart features.

The major planned milestones include:

- W4 – Project Proposal and Report Submission
- W7 – Prototype and Report Submission
- W8 – Improvement Plan
- W9–W10 – MST Week
- W11–W12 – Progress Monitoring
- W13 – Buffer
- W14 – Progress Monitoring
- W15 – Diwali Break
- W16 – Progress Monitoring over the second prototype
- W17 – Final Prototype, Presentation and Report Submission
- W18 – Assessment Week / Buffer
- W19–W20 – EST Week



Course Timeline / Academic Milestones	
Week	Course Milestone
W4	Project Proposal + Report Submission
W7	Prototype + Report Submission
W8	Improvement Plan
W9-W10	MST Week
W11-W12	Progress Monitoring
W13	Buffer
W14	Progress Monitoring
W15	Diwali Break
W16	Progress Monitoring (Improvement over Second Prototype)
W17	Final Prototype + Presentation + Report Submission
W18	Assessment Week (Buffer)
W19-W20	EST Week

The proposal follows an incremental approach: core complaint workflow first, advanced smart features later, with testing and feedback throughout development. W13 is a buffer and W15 is the Diwali break.

Figure 3: Project Gantt Chart

The schedule provides a reference for organizing implementation, testing, improvement, and final submission activities.

11. Repository Organization

The project artifacts are organized in the GitHub repository using separate folders for source material, documentation, and project journals.

The current organization includes:

SOFTWARE-ENGINEERING/

```
assets/
  activity-diagram.png
  gantt-chart.png
  use-case.png
```

```
docs/
```

```
activity-diagram.pdf
gantt-chart.pdf
use-case.pdf

frontend/

journals/
  1024170429_achintya/
  1024170437_hittin/

project_proposal/

README.md
```

The assets directory contains visual project artifacts, while the docs directory contains printable documentation versions.

The journals directory maintains individual project journals for the team members.

The repository is maintained using Git and GitHub to support collaborative development and version control.

12. Software Engineering Understanding

The requirements analysis and system modelling activities provided an understanding of how a software system can be divided into actors, functional requirements, and workflows before implementation.

The Use Case Diagram provides an understanding of *who interacts with the system and what functions they perform*.

The Activity Diagram provides an understanding of *how a complaint moves through the system from submission to closure*.

The Gantt Chart provides an understanding of *when different development activities are planned*.

Together, these artifacts provide complementary views of the same software project.

The analysis also highlighted the importance of defining responsibilities for different user roles before implementing authentication, authorization, database operations, and user interfaces.

13. Result

The following activities were completed as part of the requirements analysis and system modelling stage:

Activity	Status
System requirements analyzed	Completed
Major system actors identified	Completed
Functional requirements identified	Completed
Major use cases analyzed	Completed
Complaint workflow understood	Completed
Use Case Diagram reviewed	Completed
Activity Diagram reviewed	Completed
Project Gantt Chart reviewed	Completed
Repository organization reviewed	Completed

14. Learning / Engineering Understanding

This activity helped develop an understanding of the relationship between requirements, system modelling, and implementation.

It was observed that a clear definition of system actors and functional requirements is important before starting implementation.

The analysis of the complaint lifecycle also helped in understanding how different modules of the application will interact.

For example, complaint submission will eventually interact with authentication, database storage, location and image handling, complaint tracking, officer verification, worker assignment, notifications, and resolution verification.

The modelling activities therefore provide a foundation for the development of the frontend, backend, database, and future smart assistance modules.

15. Conclusion

Journal 02 documents the requirements analysis and system modelling activities for the Smart City Civic Complaint and Management System.

The major system actors, functional requirements, use cases, and complaint lifecycle were identified and analyzed.

The Use Case Diagram, Activity Diagram, and Gantt Chart provided different perspectives of the project by representing system functionality, process flow, and development planning respectively.

The analysis establishes a clearer understanding of the system before implementation. The next stages of development can therefore build upon the defined requirements and workflow while maintaining consistency with the established project scope.